## ELEC-237 ELECTRONICS LABORATORY-I

# Final Experiment

## Calculator opamp


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***OBJECTIVE:*** Implementing of aritmetic operators using operational amplifiers and examine ways of adding, subtracting, and integrating.




**DEPARTMENT OF ELECTRONICS ENGINEERING**

**GEBZE TECHNICAL UNIVERSITY**

# 1 Dc adder circuit with opamp

## 1.1 Some theoric

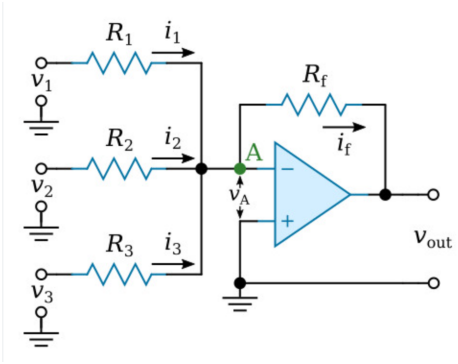



*Figure 1:Adder circuit*

This circuit work as a Adder depending on the relation between resistors. If we do KCL at Inverting node, we achieve $i_1+i_2+i_3 = i_f$. Therefore we obtain Vout formula:

$$\boldsymbol{V_{out} = -\left[\left(\frac{R_f}{R_1}\right)V_1 + \left(\frac{R_f}{R_2}\right)V_2 + \left(\frac{R_f}{R_3}\right)V_3\right]}$$

## 1.2 Experiement time

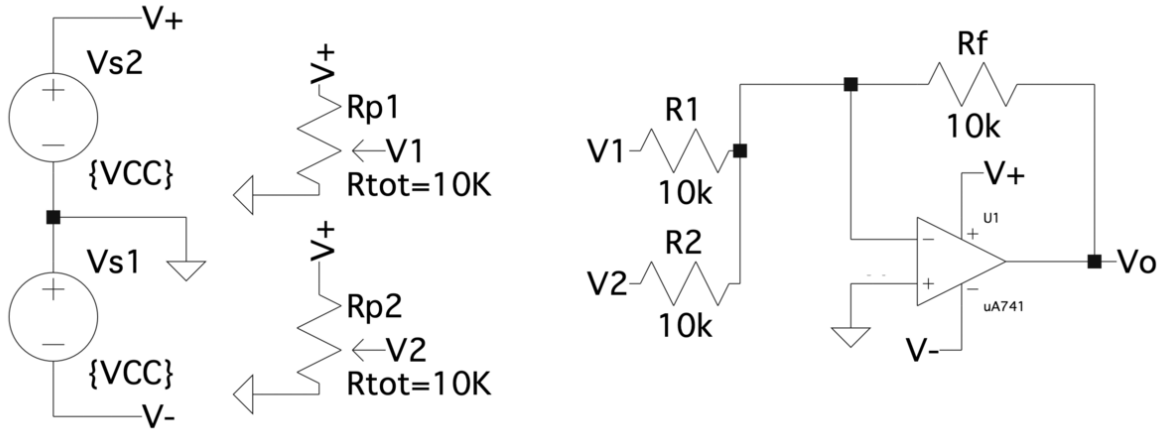



*Figure 2: DC adder circuit*

Feed the opamp with -10V +10V.Verify that this circuit work like adder observing input and output signals.

a) Set V1 to 2V, set V2 to 1 V and observe the output Vo.
b) Set V1 to 1V, set V2 to 3 V and observe the output Vo.
c) Set V1 to 4V, set V2 to 3V and observe output Vo.
d) Set V1 to 5V, set V2 to 7V and observe output Vo. Did it work successfully for this case or not why?!

Fill the Table 1 with the input and output measurements.

| V1 | V2 | Vo |
|---|---|---|
| | | |
| | | |
| | | |
| | | |

*Table 1:DC adder circuit measurements*

## 2 AC adder circuit and gain observation

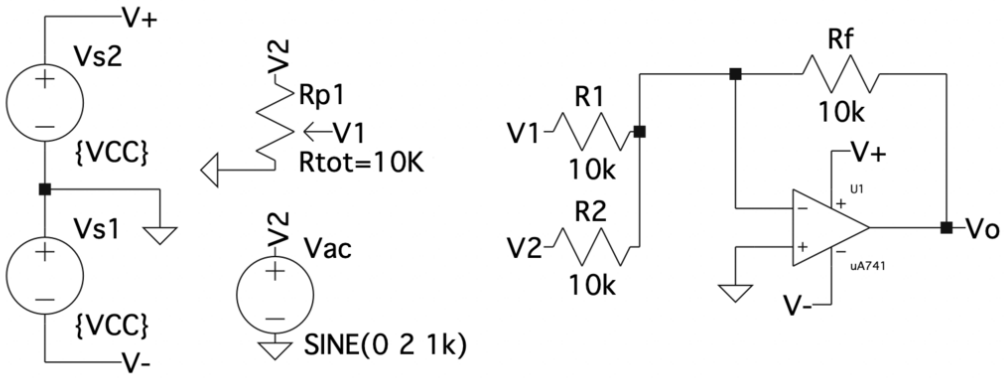



*Figure 3: AC adder circuit*

Feed the opamp with -10V +10V. Set the function generator to 3 V peak to peak with 1khz frequency. Let this be the voltage V2. Divide this voltage with a potentiometer to obtain V1 = 2 V peak to peak voltage.

a) Observe the input and output of the circuit. And measure the phase angle. Fill Table 2 with data.

| V1 | V2 | Vo | Phase |
|---|---|---|---|
| | | | |

*Table 2*

b) Shunt the resistor Rf with 10k and observe output.
c) Restore the configuration in a. Than shunt resistor R1 with 10 k and observe output.
d) Restore the configuration in a. Than shunt resistor R2 with 10 k and observe output.

Fill the data observed in b, c and d to Table 3.

| V1 | V2 | Vo |
|---|---|---|
| | | |
| | | |
| | | |

*Table 3*

e) How the gain change referred to Resistors?

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# 3 DC subtractor circuit

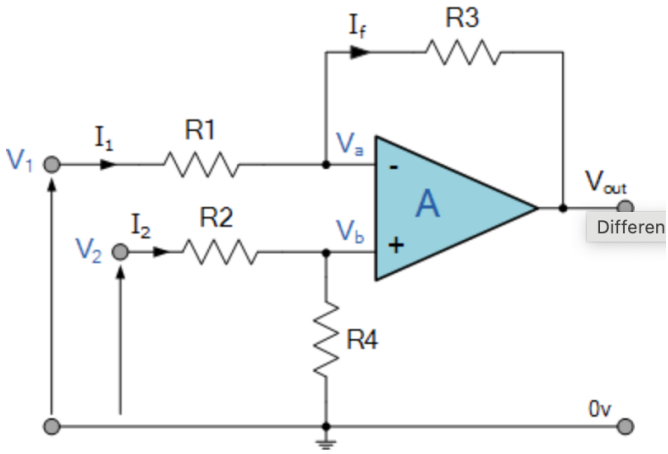



*Figure 4: subtractor amplifer*

When R1 = R2, and R3 = R4. The output voltage can be obtained by this equation using superposition theorem.

$$Vout = \left(1 + \frac{R2}{R1}\right) * Vb - \left(\frac{R2}{R1}\right) * V1$$

And

$$Vb = \left(\frac{R2}{R1 + R2}\right) * V2$$

So

$$Vout = \left(\frac{R2}{R1}\right) * V2 - \left(\frac{R2}{R1}\right) * V1$$

$$= \frac{R2}{R1}(V2 - V1)$$

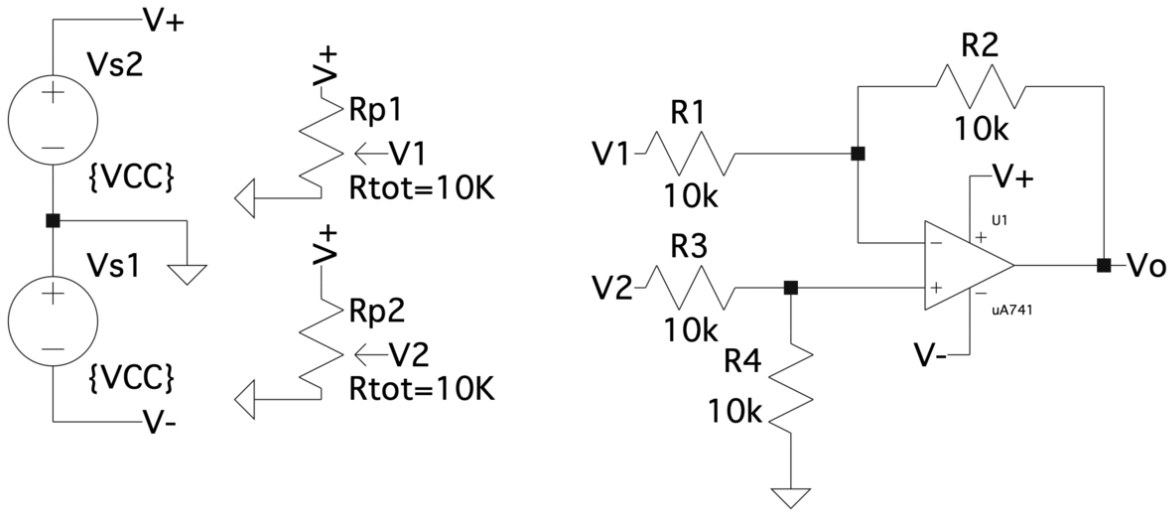



*Figure 5: DC subtractor circuit*

Feed the opamp with -10V +10V.Verify that this circuit work like subtractor (V2 – V1) observing input and output signals.

a) Set V1 to 2V, set V2 to 1 V and observe the output Vo.
b) Set V1 to 1V, set V2 to 3 V and observe the output Vo.
c) Set V1 to 4V, set V2 to 3V and observe output Vo.

Fill the Table 4 with the input and output measurements.

| V1 | V2 | Vo |
|---|---|---|
| | | |
| | | |
| | | |
| | | |

*Table 4:DC subtractor circuit measurements*

# 4 AC subtractor circuit and gain

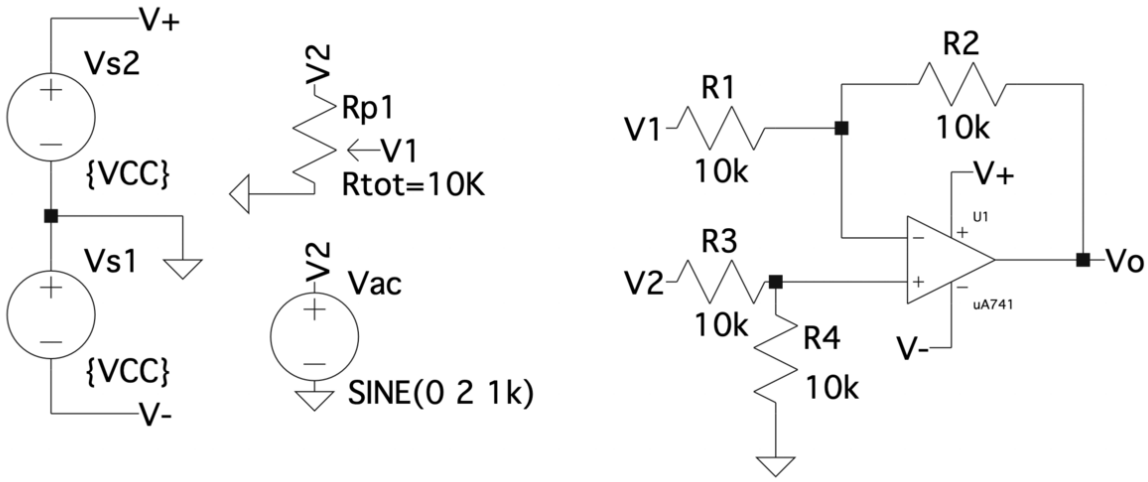



*Figure 6: AC subtractor Circuit*

Feed the opamp with -10V +10V. Set the function generator to 3 V peak to peak with 1khz frequency. Let this be the voltage V2. Divide this voltage with a potentiometer to obtain V1 = 2 V peak to peak voltage. In this circuit always R2 = R4 and R1 = R3.

a) Observe the input and output of the circuit. And measure the phase angle. Fill Table 5 with data.

| V1 | V2 | Vo | Phase |
|---|---|---|---|
| | | | |

*Table 5*

b) Shunt resistor R1 and R3 with 10 k and observe output.
c) Restore the configuration in a. Than shunt resistor R2 and R4 with 10 k and observe output.

Fill the data observed in b, c and d to Table 6.

| V1 | V2 | Vo |
|---|---|---|
| | | |
| | | |

*Table 6*

f) How the gain change referred to Resistors?

.....

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.....

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# 5 Integrator Circuit and function generator

## 5.1 Some theory

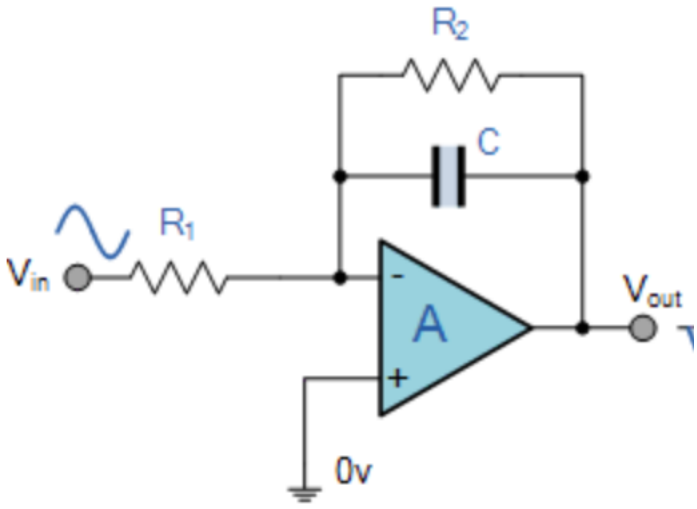
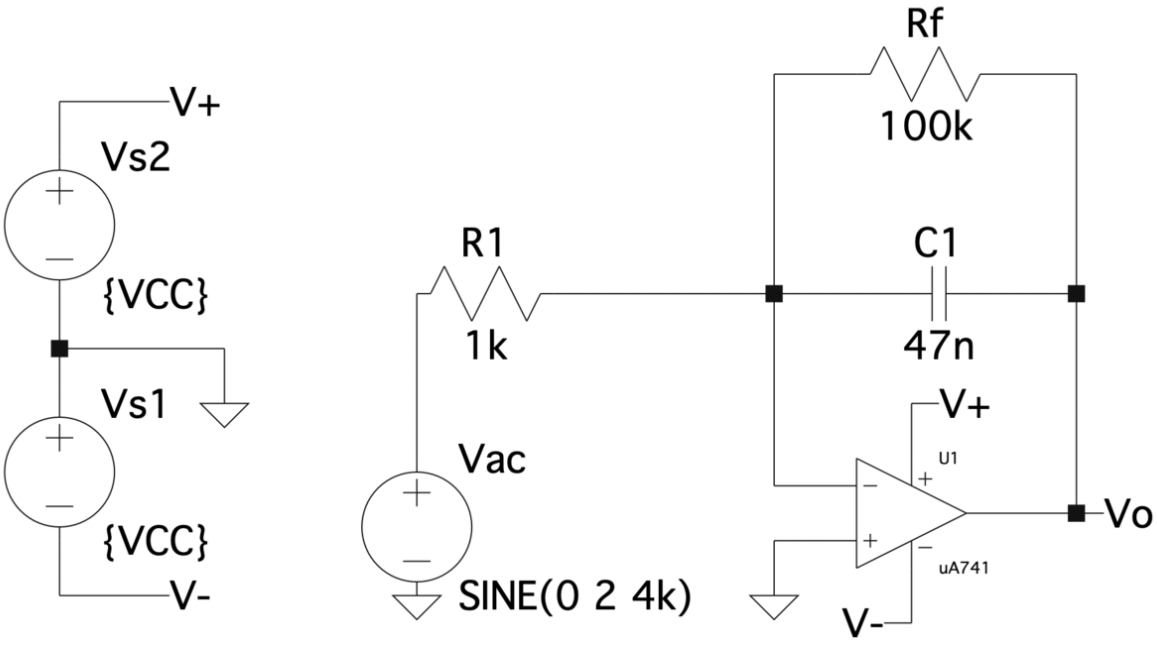



*Figure 7: Integrator circuit*

In this circuit Vx = 0 and Iin = Ic = I in ac (higher frequency than cutoff frequency), because the noniveting input of the amplifier is grounded. So from input side $\frac{Vin}{R1} = I$ and $I = Ic = C\frac{dVc}{dt} = -C\frac{dVout}{dt}$ . And from these 2 equations we obtain that $\frac{Vin}{R1} = -C\frac{dVout}{dt}$ And let's integrate the two sides to obtain the final expression. $Vout(t) = \left(-\frac{1}{R*C}\right)\int Vin(t)dt + Vout(0+)$

## 5.2 Experiement time




Feed the opamp with -10V +10V.

a) Set the function generator to sinusoidal 4 V peak to peak with 4khz frequency. Observe input and output signals and draw it.

**Waveform of Vin and Vout for sine input**

**.....V / cm**

**.....ms / cm**

**Phase Difference: .....**

b) Set the function generator to square wave with same amplitude and frequency as a. (if not work try 1khz) Observe input and output signals and draw it.



**Waveform of Vin and Vout for squarewave input**

**.....V / cm .....ms / cm**

c) Set the function generator to triangle wave with same amplitude and frequency as a. Observe input and output signals and draw it.



**Waveform of Vin and Vout for squarewave input**

**.....V / cm .....ms / cm**

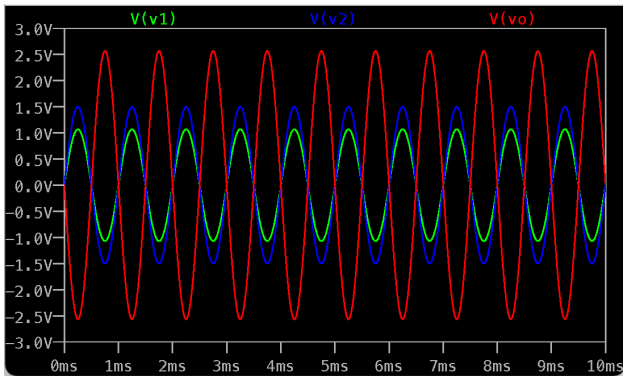
# 6 Key answer

## 6.1 Dc adder circuit with opamp

| V1 | V2 | Vo |
|---|---|---|
| 2V | 1V | -3 |
| 1V | 3V | -4 |
| 4V | 3V | -7 |
| 5V | 7V | -9.8V |

*Table 7:DC adder circuit measurements*

## 6.2 Ac adder circuit with opamp




| V1 | V2 | Vo | Phase |
|---|---|---|---|
| 2Vpp | 3Vpp | 5Vpp | 180 |

*Table 8*

| V1 | V2 | Vo |
|---|---|---|
| 2Vpp | 3Vpp | 2.5Vpp |
| 2Vpp | 3Vpp | 8 Vpp |
| 2Vpp | 3Vpp | 7 Vpp |

*Table 9*

## 6.3 DC subtractor circuit

| V1 | V2 | Vo |
|---|---|---|
| 2V | 1V | -1V |
| 1V | 3V | 2V |
| 4V | 3V | -1V |

*Table 10:DC subtractor circuit measurements*

## 6.4 AC subtractor circuit

Feed the opamp with -10V +10V. Set the function generator to 3 V peak to peak with 1khz frequency. Let this be the voltage V2. Divide this voltage with a potentiometer to obtain V1 = 2 V peak to peak voltage. In this circuit always R2 = R4 and R1 = R3.

| V1 | V2 | Vo | Phase |
|---|---|---|---|
| 2Vpp | 3Vpp | 1Vpp | 0 |

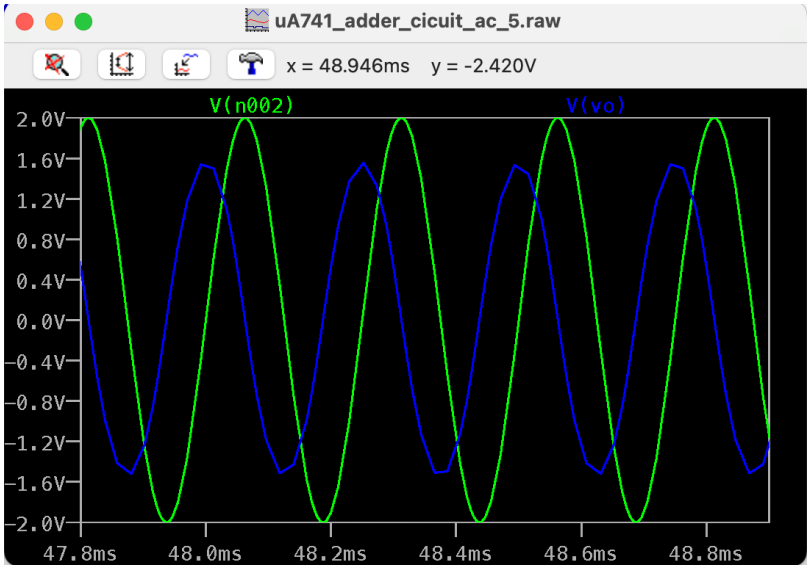
*Table 11*

| V1 | V2 | Vo |
|---|---|---|
| 2Vpp | 3Vpp | 2Vpp |
| 2Vpp | 3Vpp | 0.5Vpp |

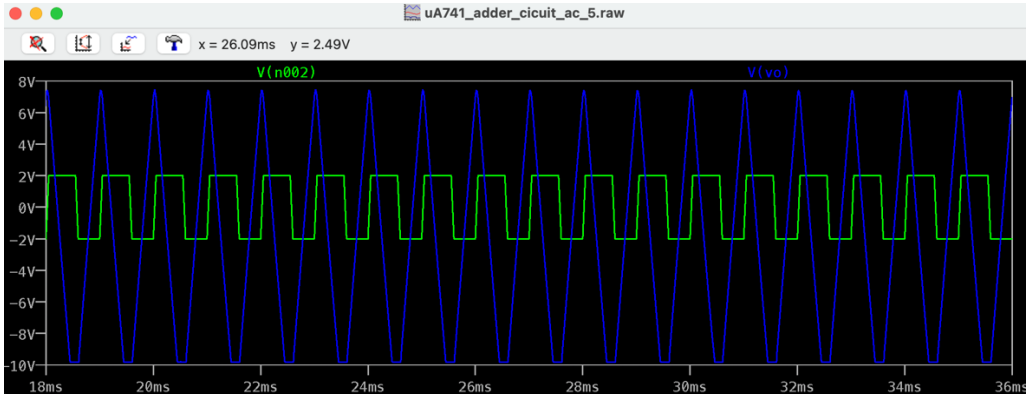
*Table 12*

## 6.5 Integrator Circuit and function generator

a)




b)




c)

